Experimental Proposal: Automated Ontology Extraction (Preliminary Prototype)

1. Objective

The primary objective of this phase is to establish a preliminary prototype to demonstrate the feasibility of an automated Ontology Extraction and Evaluation pipeline. This proof-of-concept validates the technical architecture (LangGraph workflow) and the semantic evaluation logic.

2. Experimental Setup

2.1 Model Selection Strategy

I) Extraction Layer (Generators): gemma-3-4b-it (Two instances but can be extended into more).
Role: Parallel generation of ontology candidates to introduce variation.

II) Evaluation Layer (Grader): gemma-3-27b-it.
Role: Adversarial judge to select the most coherent structure between the two candidates.

III) Structuring Layer (Formatter): gemini-2.5-flash-lite.
Role: Strict schema enforcement (Pydantic/JSON) for downstream processing.

2.2 Data Selection & Characteristics

To validate the pipeline architecture and calibrate our semantic metrics, we initially utilize a synthetically generated dataset from ChatGPT, which provided a controlled environment for feasibility testing. For the core experiment, we will transition to real-world medical reports from LMU Klinikum, shifting from clean, synthetic text to complex, unstructured clinical data to rigorously test the system's performance on authentic medical terminology and reasoning.

2.3 Evaluation Protocol (Baseline)

For this prototype, we implement a Semantic Similarity Evaluation to move beyond exact string matching. We utilize sentence-transformers/all-MiniLM-L6-v2 to embed extracted terms.

Method: A greedy matching approach with a similarity threshold (0.75) determines if the model successfully retrieved the correct concept (e.g., matching "Company" to "Corporation").

Hallucination Check: We also calculate Position Accuracy, verifying if the extracted triplet aligns with the exact source sentence cited in the ground truth.

3. Methodology Breakdown

To evaluate the performance of the AI without relying on exact word-for-word matches (which can be too strict), we use a semantic approach. This means we judge the model based on meaning, not just spelling. Here is how the four key metrics are calculated.

3.1 The Core Concept: Semantic Matching

Instead of checking if two words are identical (e.g., "Doctor" == "Physician"), we convert every word into a mathematical list of numbers called a Vector Embedding. We measure the distance (Cosine Similarity) between the Model's prediction and the Ground Truth.

The Threshold: If the distance is close enough (similarity score > 0.75), we count it as a correct match.

3.2 Metric 1 & 2: Entity and Relationship Accuracy

This measures if the model found the correct "Nouns" (Entities) and "Verbs" (Relationships).

Step 1: We take all the entities/relationships the model found (e.g., "Patient," "Doctor").
Step 2: We compare them against the official list of correct entities/relationships (Ground Truth).
Step 3: We calculate Accuracy. Example: If the text had 10 entities/relationships and the model found 6 of them (even if it used slightly different names), the accuracy is 60%.

3.3 Metric 3: Triplet Accuracy

Triplets are the building blocks of knowledge. They form a chain: (Subject > Relationship > Object).

This metric is much harder to satisfy because it requires a perfect chain. For a triplet to be counted as correct, all three components must match - The Subject must be semantically similar (e.g., "Patient"). The Relationship must be semantically similar (e.g., "took"). The Object must be semantically similar (e.g., "Medication").

If the model gets the entities right but the relationship wrong (e.g., "Patient prescribed Medication" instead of "took"), the entire triplet is marked as Incorrect.

3.4 Metric 4: Position Accuracy (The Hallucination Check)

This checks whether the model found the triplet at the correct location or not. It is crucial for preventing hallucinations (where the AI makes up facts that aren't in the text).

The Logic: Every time the model claims it found a fact (a Triplet), we force it to quote the exact Source Sentence from the original text where it found that fact.

The Check: We compare the model's quoted sentence against the official source sentence for that fact.

The Scoring: If the model pointed to the correct sentence, we give it a score of 1 otherwise 0. A mean score of 62% means that 62% of the triplets were correctly predicted at the exact location.

4. **Preliminary Results (Prototype Run)**

We conducted an initial feasibility run using the synthetic dataset to establish a performance baseline. The most promising metric was Position Accuracy, which reached 61.9%. This indicates that the model is generally effective at "grounding" its extraction, when it identifies the triplet, it correctly cites the specific source sentence in the text, showing low hallucination regarding the location of data.

In terms of semantic content, the model performed moderately well on Relationships with an accuracy of 57.5%, suggesting it can reliably identify the interaction verbs between concepts (e.g., distinguishing "treats" from "causes"). However, performance dropped for full Triplets (37.5%) and Entity Classes (31.4%). The lower Entity Class score likely reflects a granularity mismatch, where the model extracts a valid but slightly different term than the ground truth (e.g., "AI Tool" vs. "Software") which naturally drags down the score for the full triplets that depend on those entities. These results represent a raw baseline without advanced prompt optimization and highlight specific targets for the next iteration of refinement.

5. **Future Evolution & Roadmap**

This current iteration is a baseline implementation. Following the successful validation of this prototype, we plan to evolve the system in the following areas:

5.1 Pipeline Robustness: Moving from simple linear extraction to iterative refinement (e.g., adding a "Reflection" node where the model critiques its own ontology before finalizing).

5.2 Data Transition: Replacing synthetic ChatGPT data with domain-specific, human-annotated "Gold Standard" datasets to test real-world performance.

5.3 Advanced Evaluation: We will refine the evaluation criteria.